

Multilayer Hemostatic Bandage Design for Rapid Clotting and Atraumatic Base-Layer Detachment

Executive summary

The strongest overall design logic is to **separate the functions of hemostasis, timed release, and structural support into different layers** rather than asking one material to do all three. The evidence base is most mature for wound-contact hemostatic layers built around **kaolin, chitosan, cellulose derivatives, or combinations of these**, while the most credible routes to non-traumatic removal use **sacrificial or trigger-detachable hydrogel interfaces** based on **PVP/PAA complexes, PEG-containing dissolvable chemistries, alginate-type ionic gels, or shear-responsive supramolecular networks**. A practical product should therefore be conceived as a **three-layer system**: a porous hemostatic wound-contact layer that concentrates blood cells and activates clotting; a wet-activated intermediate layer that **starts strong enough to tolerate manual compression** but then loses interlayer adhesion after a defined dwell time; and a breathable, conformable outer support layer with a skin adhesive selected for atraumatic periwound wear. ¹

For a **first-generation, lowest-regulatory-risk product**, the best lead concept is a **physical-action hemostatic bandage** built from a **kaolin + carboxymethyl chitosan or kaolin + chitosan/cellulose composite wound-facing layer**, a **sacrificial PVP/PEG or PVP/alginate release film**, and a **polyurethane film or foam-backed outer layer with soft-silicone periwound adhesive**. This avoids biologics, preserves a plausible U.S. 510(k)-style device path for an external topical wound dressing without thrombin or other biologics, and uses materials that already have strong preclinical and translational precedent. In contrast, **thrombin- or fibrin-containing concepts** can likely improve clotting kinetics but substantially increase formulation, sterilization, shelf-life, and regulatory complexity, especially in the EU where medicinal substances and human/animal-derived derivatives can push classification to **Class III**. ²

The most important **compatibility lesson** is that some “good hemostats” should **not** be co-located. In particular, **oxidized regenerated cellulose** creates a low-pH environment; official SURGICEL information states that this low pH **destroys thrombin activity**, so an ORC-containing wound-contact layer should not be paired in the same microenvironment with thrombin if the design intent is to preserve thrombin bioactivity. That makes ORC attractive for a physical hemostatic/contact layer, but not for a thrombin-loaded one. ³

A reasonable initial target product profile for an unconstrained external-use bandage is: **initial hemostasis within 3–5 minutes under manual compression, bench clotting time at or below roughly 1–2 minutes in recalcified whole-blood assays, interlayer peel stable during the first compression interval**, and then **at least a 70–80% reduction in base-layer peel/load after clot maturation**, with no rebleeding on removal in staged ex vivo and animal testing. Those numeric targets are best treated as **engineering stage-gates**, not as fixed regulatory thresholds; they are inferred from current hemostatic dressing performance, detachable-adhesive literature, and standard tissue-adhesive / peel-test frameworks. ⁴

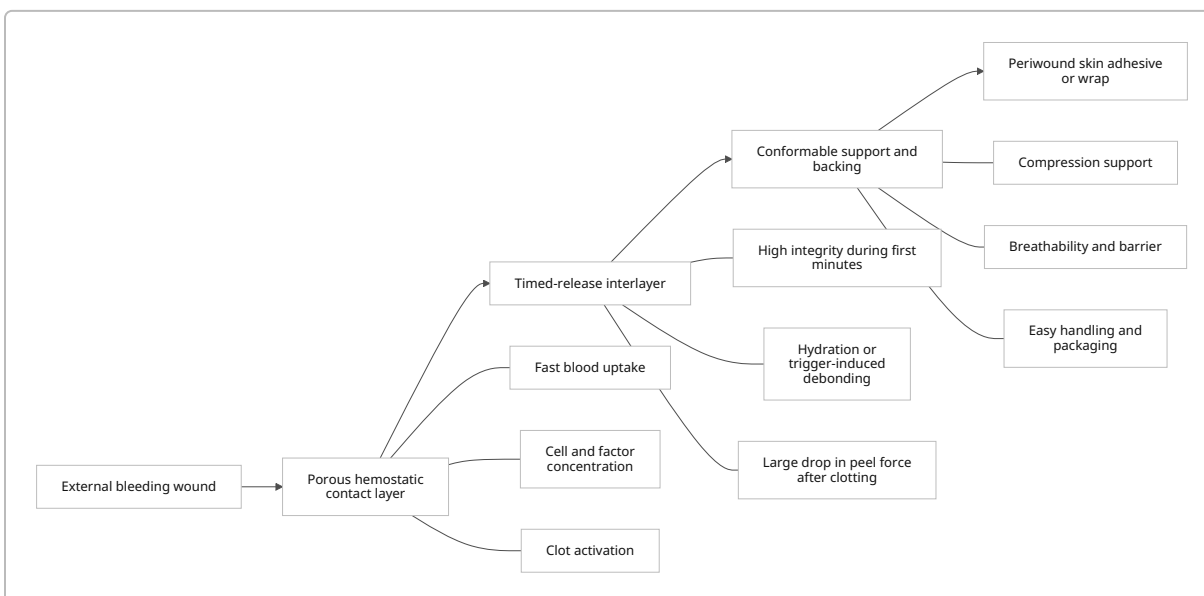
The recommended development sequence is to **start with a non-biologic lead**, tune release timing with benchtop blood-contact and peel testing, then only consider a thrombin/fibrin variant if the physical-action product cannot reach the desired clotting speed. That prioritization minimizes the two hardest translational risks: **protein stability during sterilization and storage**, and **up-classification into higher-burden regulatory pathways**. ⁵

Design objectives and translational boundary

Because no wound type, population, or setting was specified, the most defensible baseline assumption is an **external, compressible acute bleeding indication** in civilian, prehospital, or hospital care rather than noncompressible internal hemorrhage. FDA's current product classification entry for a **hemostatic wound dressing without thrombin or other biologics** defines it as a device intended **to temporarily control bleeding and cover external wounds**, and emphasizes predominantly physical mechanisms such as absorption and tamponade. That is the right reference frame for a multilayer detachable bandage. ⁶

Within that boundary, the product has to satisfy four coupled requirements. First, the wound surface must quickly accumulate platelets, red cells, fibrin, or all three. Second, the bandage must maintain enough cohesion and interlayer integrity to survive packing or compression during the early clotting window. Third, once a stable clot has formed, the **outer/base portion** should detach without pulling away the immature clot or re-opening the wound. Fourth, the whole stack must remain manufacturable, sterilizable, and stable over a commercial shelf life. These are not separate problems: choices that maximize clotting often increase adhesion or residue, and choices that maximize removability can reduce hemostatic efficiency. ⁷

A useful way to phrase the design problem is therefore:



That architecture aligns well with current wound dressing and hemostat classification logic in both the U.S. and EU: the more the product depends on **physical action**, the more tractable the pathway; the more it depends on **licensed thrombin, fibrinogen, or other ancillary medicinal/biologic action**, the heavier the burden becomes. ⁸

Layer-by-layer evidence review

Wound-facing hemostatic layer

The wound-contact layer should do three things simultaneously: **take up fluid fast, concentrate formed elements**, and **accelerate coagulation by either physical or biologic means**. The strongest choices in today's literature are not single "magic" materials but **porous architectures** built from minerals plus polymers, or biopolymers plus controlled pore structure. Whole-blood clotting, platelet adhesion, and reduction in time-to-hemostasis all improve when scaffold porosity, swelling, and hemostatic chemistry are tuned together. ⁹

Kaolin remains one of the most translationally credible hemostatic actives for an external bandage. FDA-cleared kaolin dressings already exist, and official product information describes kaolin-bound gauze as an external hemostatic dressing used with direct pressure. Preclinical literature and TCCC-oriented analyses show that kaolin gauze is effective, and chitosan-based gauzes have often performed comparably in swine and other models. Kaolin is attractive because it is inexpensive, dry-stable, and compatible with terminally sterilized physical dressings. Its main limitation in this specific project is that a mineral-loaded face layer can leave residue and can stiffen or abrade the interface if the loading is excessive. ¹⁰

Chitosan is especially attractive when the design goal includes both hemostasis and a controlled handoff to atraumatic removal. Its cationic nature supports red-cell and platelet interactions, and a large modern literature shows effective chitosan sponges, membranes, nanofibers, and composite hemostats. Chitosan also combines well with cellulose nanofibers and carboxymethyl chitosan architectures that expand, absorb blood, and improve hemostasis in deep or penetrating injury models. The main engineering risks are batch variability, moisture sensitivity, and sensitivity of some chitosan structures to radiation sterilization. ¹¹

Thrombin and **fibrin/fibrinogen-thrombin systems** can reduce clotting time further because they directly leverage the final coagulation cascade. FDA labeling for topical recombinant thrombin and fibrin sealants confirms that these materials are supplied as sterile biologic products for topical hemostatic use, and preclinical dressing studies show strong efficacy when thrombin or fibrinogen are captured in sponges or electrospun matrices. The problem is not efficacy; it is translational friction. These agents create a much harder sterilization and storage problem and often move the product toward a higher-burden device/biologic or combination pathway. ¹²

Oxidized cellulose / oxidized regenerated cellulose is useful when the goal is a highly absorbent, clot-supportive surface that can also swell into a gelatinous scaffold. Official SURGICEL information describes ORC as a physical hemostat that swells after saturation and supports platelet adhesion/aggregation, but also warns about excess retention, migration, and foreign-body complications. Critically, official product information also states that ORC's low pH destroys thrombin activity, which makes ORC **poorly suited to co-formulation with thrombin** in the same active layer. That compatibility constraint is highly actionable for multilayer design. ¹³

Zeolites have recognized hemostatic value as highly absorbent mineral systems, and they remain important comparators in the overall hemostatic-agent landscape, but for a detachable multilayer skin bandage they are less attractive than kaolin/chitosan/cellulose routes. The main reasons are practical: particulate mineral loading, wound-bed residue management, and a weaker fit with the goal of a gentle

contact/removal interface. For this product brief, zeolites are better treated as a benchmark family than as a lead material family. ¹⁴

The highest-value form factors for this application are **porous sponges/foams** and **electrospun fiber mats**, with **hydrogels** used more selectively. Sponges excel at absorption and tamponade, electrospun mats excel at large surface area and rapid wetting, and hydrogels excel at conformity but can become too adhesive unless intentionally designed for debonding. ¹⁵

Wound-facing option	Mechanism and strengths	Important liabilities	Verdict for this project
Kaolin in gauze/fiber/sponge	Strong translational precedent; physical-action hemostasis; dry stability; easy integration into terminally sterilized dressings. ¹⁶	Particulate residue if over-loaded; can stiffen interface.	Lead option
Chitosan or CMCS sponge/fiber	Strong cell interaction, good swelling/absorption, broad composite literature. ¹⁷	Radiation sensitivity for some formats; raw-material variability.	Lead option
Kaolin + chitosan composite	Synergistic physical + polymeric hemostasis; strong preclinical data. ¹⁸	Formulation complexity; mineral dispersion control needed.	Best first-gen chemistry
Thrombin-loaded sponge/fiber	Fastest clotting potential; strong active hemostasis. ¹⁹	Protein stability, aseptic loading, higher regulatory burden.	Second-gen / hospital variant
Fibrin or fibrinogen-thrombin patch	Direct fibrin clot formation; proven surgical hemostasis. ²⁰	Highest complexity and cost; cold-chain / reconstitution issues for some systems.	Not first choice
ORC or oxidized cellulose composite	Swelling, platelet-supportive surface, absorbable architectures. ²¹	Excess retention risks; incompatible with thrombin in same acidic microenvironment. ²²	Useful non-biologic alternative
Zeolite-based layer	Strong absorbency and historic hemostatic use. ¹⁴	Less favorable fit with atraumatic detachable interface.	Comparator, not lead

Intermediate release layer

This layer is the intellectual center of the product. It has to hold the stack together during the first minutes of pressure and blood uptake, then **lose adhesion or cohesion in a controlled way** once the clot is stable enough to survive base-layer removal. The most credible mechanisms are not classic “low-stick release liners,” because this interface must operate **wet, under blood, and on a living wound**. The better analogy

comes from **wet-tissue adhesives that can later be detached, biodissolvable hydrogel complexes, and moisture-triggered sacrificial substrates.** ²³

A very practical route is a **PVP-containing sacrificial interface**. PAA/PVP complex films have been shown to form strongly bioadhesive hydrogels on wet tissue and then slowly dissociate/dissolve, while a dual-layer microneedle wound dressing used a **PVP backing substrate that dissolved within minutes** in a moist local environment. That is highly relevant to a detachable base-layer concept: PVP is cheap, processable, and can be used as a hydrated “failure plane” if its thickness and formulation are tuned carefully. ²⁴

PEG-based dissolvable or trigger-cleavable interfaces are also compelling, especially for higher-end products. A well-known on-demand dissolvable burn dressing used a PEG-based crosslinker in a dendritic hydrogel that could be dissolved by thiol-thioester exchange, and several 2022–2024 tissue-adhesive papers show strong wet adhesion followed by water-, shear-, chemical-, or temperature-triggered debonding. The limitation is translational simplicity: these approaches often need an external trigger or a more complex chemistry package than a standard bandage ideally should. ²⁵

Alginate is attractive because it can form soft ionic hydrogels and can be destabilized by ion exchange or chelation. The literature confirms that polymerized alginate can be dissolved by chelation, although high chelator concentrations are not desirable at the wound interface. In practice, alginate is most attractive here as part of a **PVP/alginate or PEG/alginate water-uptake interface** whose adhesion drops through swelling and ionic relaxation rather than through deliberate addition of EDTA or citrate at the bedside. ²⁶

Shear-thinning or thixotropic layers can be engineered so that non-covalent interactions collapse under peel or shear, enabling easier removal. That principle is elegant and well supported in detachable bioadhesive literature, but it fits best when the interlayer is a **contained hydrogel film** or when the product is an injectable/sealant application. For a dry-packaged bandage, it is more complex to implement robustly than a sacrificial, water-swelling PVP/PEG interface. ²⁷

The most robust first-generation release-layer concept is therefore a **thin, moisture-activated sacrificial hydrogel film** built from **PVP with either PEG or a low proportion of alginate**, optionally lightly crosslinked or complexed so that it is strong enough during the first 2–3 minutes of compression but undergoes a major peel-force drop by about 5–15 minutes after blood exposure. ²⁸

Release mechanism	Why it works	Main translational issue	Best use here
PVP-rich sacrificial film	Water uptake and dissolution can sharply reduce adhesion after wetting; precedent in moist-environment detachable dressings. ²⁴	Too much PVP can fail too early.	Best first-gen option
PEG-containing dissolvable interface	Good processability; can support controlled hydration or trigger-cleavage chemistry. ²⁹	More chemistry and trigger-validation burden.	Good backup option

Release mechanism	Why it works	Main translational issue	Best use here
Alginate-containing ionic gel	Soft, hydrophilic, tunable, ionic relaxation and swelling can reduce interlayer strength over time. ²⁶	Chelator-trigger risks if over-engineered.	Useful co-component
Shear-thinning / thixotropic layer	Debonds under peel/shear by breaking reversible bonds. ²⁷	More difficult to package as a standard dry bandage.	Advanced variant
Stimuli-responsive wet adhesive	Can offer very large adhesion changes on demand. ³⁰	User-applied trigger may complicate workflow and labeling.	Niche / premium variant

Base and outer support layer

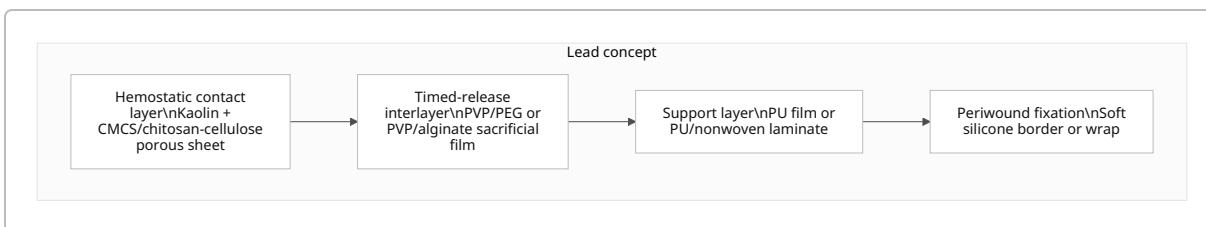
The outer layer should be treated as a **mechanical compression and handling structure**, not as the main hemostatic actor. Polyurethane-based wound-dressing literature consistently supports PU films and foams as good support/backing candidates because they are conformable, breathable, and capable of moisture-vapor transmission while also serving as barriers. Contemporary reviews also note that PU layers can be gas-permeable and suitable as wound-dressing backings. A WVTR in the low-thousands of g/m²/24 h is a reasonable design region for breathable outer films; one recent breathable composite wound dressing reported **4.83 × 10³ g/m²·24 h**, which is directionally useful as a benchmark. ³¹

For **skin attachment**, the evidence most strongly favors **soft silicone adhesives** when atraumatic removal and patient comfort matter. Clinical studies and dressing evaluations report painless or atraumatic removal and improved comfort with silicone adhesive layers and soft-silicone contact systems. For this project, that suggests using silicone **around the periwound** or on an optional border, while keeping the hemostatic/contact stack itself largely free of skin PSA chemistry. Acrylic PSAs can provide higher hold but are more likely to increase skin-trauma risk, especially if the bandage is expected to stay on for hours. ³²

For high-bleeding scenarios, the backing should probably be an **elastic film-plus-nonwoven** or **film-plus-thin-foam** laminate rather than a film alone, because the outer layer must transmit manual compression without wrinkling, bunching, or shear concentration. The low-cost, high-volume solution is therefore not exotic: a PU film or laminated PU/nonwoven support, plus a separate border or wrap-based fixation strategy. ³³

Candidate multilayer concepts and manufacturing routes

The most promising multilayer structures can be summarized as follows.



Recommended lead concepts

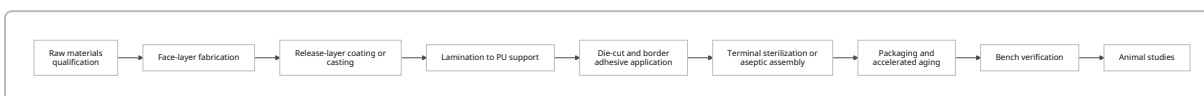
Candidate stack	Suggested construction	Why it is attractive	Expected stage-gate performance
Kaolin + CMCS/ chitosan-cellulose lead	Electrospun or freeze-dried wound-contact layer with kaolin dispersed in CMCS/ chitosan-cellulose matrix; thin PVP/PEG or PVP/ alginate release film; PU backing with silicone border. ³⁴	Best combination of hemostatic precedent, cost, dry stability, and manageable regulation.	Target bench clotting ≤ 120 s; hemostasis under compression $\leq 3\text{--}5$ min; interlayer peel drop $\geq 70\%$ by 10 min. <i>Engineering target inferred from current literature.</i> ³⁵
ORC + CMCS alternative	ORC/oxidized-cellulose-chitosan sponge on sacrificial PVP release layer and PU support. ³⁶	Strong absorbency and physical hemostasis; good if residue/ absorbability is prioritized.	Similar clotting target to lead, but slightly more caution on retained-material effects and removal of excess. ³⁷
Thrombin-loaded high-performance variant	TOCN/gelatin/thrombin or thrombin-loaded electrospun nanofiber face layer; PEG-cleavable or biodissolvable release interface; PU support. ³⁸	Fastest clotting likely achievable.	Target bench clotting ≤ 90 s and strong early hemostasis, but only if aseptic loading and biologic stability are solved. ³⁹
Janus self-pumping concept	Fiber face plus kaolin/ Ca^{2+} hydrogel layer assembled as a directional-drainage dressing. ⁴⁰	Highly relevant to the “rapid clotting + easy clot release” design goal.	Promising, but manufacturing and interlayer-control complexity are higher than for the lead. ⁴¹

The **preferred first prototype family** is the first row, with three subvariants differing mainly in release-layer composition and thickness. That lets the program tune the most uncertain parameter — the timing of adhesion reduction — without changing the wound-contact chemistry too early. ⁴²

Manufacturing fit

The favored manufacturing sequence is: **form the hemostatic face layer, cast or coat the sacrificial release layer, laminate to a PU support**, then **attach a perimeter adhesive border if needed**. For a chitosan/cellulose/kaolin face, the most plausible scalable routes are **freeze-drying** for sponge structures or **electrospinning** for nanofibrous mats. Electrospinning is especially attractive because industrial-scale and needleless approaches already exist, and commercial biomedical electrospun products are no longer hypothetical. By contrast, **3D printing** is excellent for prototyping and wound-shape customization, but current literature still treats scale-up and throughput as major challenges for commercial volume manufacturing. ⁴³

A realistic first manufacturing flow is:



In cost terms, **mineral + polysaccharide + PU** is the most favorable family. Kaolin, PVP, PVA/CMCS/chitosan, and PU film are all inherently less expensive and easier to source than licensed thrombin/fibrin systems. Protein-loaded products also incur additional costs from low-moisture processing, aseptic handling, potency testing, and likely more complex release/stability studies. ⁴⁴

Verification strategy and prioritized experimental plan

Bench and in vitro testing

A rigorous bench program should explicitly test both hemostasis and **timed debonding**. The best starting assays are:

1. **Whole-blood clotting / coagulation index assays** using fresh citrated blood with recalcification and comparison to gauze and a commercial positive control. Jesty and colleagues described a useful in vitro hemostatic-index method based on measuring flow of recalcified platelet-rich plasma through dressing samples; separate clotting-time and clotting-index approaches are also widely used across contemporary hemostatic-material studies. ⁴⁵
2. **Platelet adhesion, RBC adsorption, and hemolysis** according to a hemocompatibility plan grounded in ISO 10993-4 and material-screening practice. A hemolysis rate below **5%** is a common screening criterion in the hemostatic biomaterials literature and fits ISO-style hemocompatibility expectations. ⁴⁶
3. **Cytotoxicity and extract testing** under ISO 10993-5, with endpoint selection under the ISO 10993-1 risk-management framework. ⁴⁷
4. **Pre- and post-clot interlayer peel/shear testing** using porcine skin or blood-conditioned soft-tissue substrates. ASTM F2255 and F2256 are suitable for soft-tissue lap-shear and T-peel comparisons, while ASTM D3330 is appropriate for the periwound pressure-sensitive border adhesive. ASTM E96/E96M should be used for WVTR of the support layer. ⁴⁸

A practical acceptance table for the first design cycle is below.

Test	Proposed protocol	Acceptance criterion
Recalcified whole-blood clotting time	Human or porcine citrated blood; compare against gauze and commercial comparator. ⁴⁹	≤ 120 s for lead non-biologic concept; ≤ 90 s stretch goal.
Hemostatic index / flow-through	Jesty-style PRP flow/hemostatic index method. ⁵⁰	Statistically superior to plain gauze; noninferior or superior to selected comparator.
Hemolysis	ISO 10993-4-aligned extract and direct-contact testing. ⁴⁶	$< 5\%$ hemolysis.
Cytotoxicity	ISO 10993-5 extract cytotoxicity. ⁵¹	$\geq 70\%$ cell viability versus control.
Interlayer peel before clotting	ASTM F2256-adapted, wet and compressed for 2 min. ⁵²	Strong enough to avoid early delamination during manual compression; target > 1.0 N/cm.
Interlayer peel after clotting	Same test after 10 min blood exposure. ⁵²	Target < 0.3 N/cm or $\geq 70\%$ peel reduction vs pre-clot.
Border adhesive peel	ASTM D3330 on skin analog / porcine skin. ⁵³	Stable fixation without skin stripping; optimize for atraumatic silicone removal.
WVTR	ASTM E96/E96M. ⁵⁴	In breathable-dressing range; avoid occlusive backing unless clinically intended.

These numeric criteria are **recommended development gates**, not codified standards. Their purpose is to force an honest go/no-go on the central claim: “rapid clotting at the wound surface, but easy detachment of the base layer after clot formation.”

Animal models

Animal work should proceed from fast screening to translational confirmation. A sensible sequence is:

Early screening

rat tail amputation, rat ear punch, or rat liver punch/biopsy models to down-select chemistry and release timing quickly. Many contemporary hemostatic-material papers use rat liver injury because it is inexpensive and sensitive to differences in clotting speed and blood loss. ⁵⁵

Intermediate confirmation

rabbit femoral artery/vein models or similar arterial injury models, especially if comparing to commercial hemostatic gauze. Kaolin/chitosan composite literature has already shown meaningful discrimination between experimental materials, conventional gauze, and QuikClot Combat Gauze in rat and rabbit femoral models. ¹⁸

Translational large-animal work

swine femoral or junctional hemorrhage with standardized compression/resuscitation, because swine remain the most relevant preclinical large-animal model for external traumatic hemorrhage control. Combat-gauze and comparator dressing studies use swine extensively, including coagulopathic and severe hemorrhage settings. ⁵⁶

For this specific product, animal testing should add one endpoint that many hemostat studies omit: **rebleeding after staged removal of the base layer**. The proper comparison is not merely time-to-hemostasis; it is **time-to-hemostasis plus atraumatic detachment without clot disruption**.

Prioritized development plan

The highest-value experimental sequence is:

Program one

build 9–12 non-biologic prototypes varying only (a) kaolin loading, (b) chitosan/CMCS-to-cellulose ratio, and (c) PVP/PEG vs PVP/alginate release-layer thickness. Do not add antimicrobial drugs, sensing functions, or therapeutic agents in this round. ⁵⁷

Program two

select the top 2–3 prototypes based on clotting time, hemostatic index, and post-clot peel reduction. Advance only prototypes that meet hemolysis and cytotoxicity gates. ⁵⁸

Program three

run ex vivo porcine-skin bleeding and removal simulations, then rat/rabbit testing. The primary success criterion is **no rebleeding on base-layer removal after a defined dwell time**, not just lower blood loss during application. ⁵⁹

Program four

only if the best non-biologic product fails the speed target, formulate a thrombin-loaded hospital variant using aseptic loading and a separate stability workstream. ⁶⁰

Regulatory, sterilization, shelf life, scale-up, and cost

In the **United States**, the cleanest pathway is the one that stays within the logic of FDA's current device classification for **hemostatic wound dressing without thrombin or other biologics** for external wounds. FDA's current product classification database lists this device type under **QSY**, while the with-thrombin/or-other-biologics device type is listed under **QSX**. FDA's 2022 advisory-panel materials also identified the main risks for these dressings — uncontrolled bleeding, infection, adverse tissue reaction, delayed healing, pathogen transmission, immunologic reaction, microbial growth, antimicrobial-resistance contribution, and retained-device foreign-body reaction — and the panel was comfortable with Class II classification logic for topical hemostatic wound dressings. A first-generation non-biologic multilayer dressing therefore has a substantially clearer device case than a thrombin/fibrin product. ⁶¹

In the **European Union**, the picture is more claim- and composition-dependent. Under MDR Rule 4, non-invasive devices contacting injured skin can range from Class I to IIa or IIb depending on whether they act mainly as a barrier/compression/absorption system, manage the wound microenvironment, or are intended for deeper injuries healing by secondary intent. But the classification escalates sharply if the device incorporates a medicinal substance with ancillary action (**Rule 14, Class III**) or is manufactured using non-viable human/animal tissues or derivatives (**Rule 18, generally Class III unless only contacting intact**

skin). That makes thrombin, fibrinogen/fibrin, and some tissue-derived components strategically unattractive for a first-market EU launch unless the performance gain is overwhelming. ⁶²

For **sterilization**, the default recommendation for the non-biologic lead is to evaluate both **EtO** and **radiation**. ISO 11135 governs EO sterilization, while ISO 11137 covers radiation sterilization. Radiation is operationally attractive, but published work shows that gamma irradiation can negatively affect chitosan membrane morphology and some material properties, so dose mapping and post-sterilization performance testing are essential. EO may better preserve polysaccharide architectures but introduces residuals and aeration burdens. ⁶³

For **protein-loaded versions**, the evidence points in the opposite direction: topical thrombin and fibrin sealants are supplied as sterile biologic products, often lyophilized or reconstituted close to use, which strongly suggests that post-loading terminal sterilization of the final protein-bearing bandage would be a major risk to potency and shelf life. In practice, a thrombin/fibrin variant should be assumed to require **aseptic integration or sterile component assembly** unless proved otherwise. ⁶⁴

For **packaging and shelf life**, ISO 11607 should frame sterile-barrier development, and ASTM F1980 should be used for accelerated aging with concurrent real-time aging. FDA-cleared dressing summaries routinely use this framework, and three-year shelf lives are common in cleared dry dressings when the chemistry is stable. QuikClot's IFU also underscores the importance of dry storage and temperature control for external hemostatic dressings. ⁶⁵

From a **manufacturing cost** perspective, the ranking is straightforward. Lowest cost and easiest scale-up: **kaolin/chitosan or kaolin/cellulose physical hemostats with coated sacrificial release films**. Moderate cost: **ORC/chitosan composite structures**. Highest cost and most difficult CMC/regulatory package: **thrombin/fibrin-loaded systems**. Electrospinning is now scale-relevant, but if the same performance can be reached with a freeze-dried sponge and coated film, the sponge route may still be preferable on cost and throughput grounds. 3D printing should remain a prototyping or niche-customization tool unless the business case specifically depends on personalized geometry. ⁶⁶

Recommendation, risk analysis, and open questions

The **recommended lead product** is:

Wound-facing layer

porous **kaolin + carboxymethyl chitosan/chitosan-cellulose** sheet, preferably as a thin sponge or nanofiber-reinforced sponge rather than loose particles. ⁶⁷

Intermediate release layer

PVP/PEG or **PVP/alginate** sacrificial film tuned to retain interlayer integrity during the first minutes of compression and then undergo a large adhesion drop once fully hydrated by blood/serum. ⁶⁸

Base/outer layer

PU film or PU/nonwoven laminate, with **soft silicone periwound adhesive** if a bordered dressing is needed; otherwise pair with a wrap or secondary fixation to keep wound-bed chemistry and skin PSA functions separate. ⁶⁹

That configuration best reconciles efficacy, removal, cost, sterilization, and regulatory practicality.

The main risks and mitigations are summarized below.

Risk	Why it matters	Mitigation
Early interlayer failure	Bandage separates during compression before clot forms.	Increase release-layer cohesion, reduce hydrophile fraction, add microperforated mechanical tie points, and verify wet peel at 2 minutes.
Late over-adhesion and traumatic removal	Base removal reopens wound.	Tune PVP fraction and film thickness; incorporate reversible hydration-driven failure plane; measure peel after blood conditioning.
Residual particulate shed	Kaolin or mineral residue complicates irrigation.	Use firmly bound mineral in polymer matrix; test particle shed under simulated use.
ORC-thrombin incompatibility	Low pH inactivates thrombin.	Do not co-formulate ORC with thrombin in same active layer. ⁷⁰
Chitosan sterilization damage	Radiation can alter structure/performance.	Compare EtO and radiation early; lock sterilization method before design freeze. ⁷¹
Biologic potency loss	Thrombin/fibrin variant may lose activity during manufacture or storage.	Keep biologics off the critical path; if pursued, run a separate aseptic program. ⁷²
Regulatory escalation	Thrombin/fibrin can trigger higher-burden pathways.	Launch first with non-biologic physical-action chemistry. ⁷³

Open questions and limitations

The literature is strongest on **hemostasis** and much thinner on **base-layer detachment after clot maturation** in trauma-style dressings. The detachable-adhesive and dissolvable-hydrogel literature provides strong mechanistic support, but direct head-to-head evidence for this exact bandage use case is still limited. That is why the experimental plan above puts unusual weight on **post-clot removal and rebleeding** endpoints. ⁷⁴

The second key limitation is that the brief did not specify whether the product should be optimized for **minor-to-moderate external bleeding, high-flow junctional trauma, or surgical oozing / post-procedural bleeding**. One platform can span these settings, but not without tradeoffs in thickness, compression mechanics, and allowable residue. The recommended lead architecture is best viewed as a **platform** that can be tuned once the ultimate indication is narrowed. ⁷⁵

Overall, if the goal is a realistic product that can move from concept to prototype to regulated device, the best answer is **not** the most biochemically aggressive dressing. It is a **physical hemostatic contact layer plus a deliberately engineered sacrificial wet interface**. That is the configuration most likely to achieve the combination the brief asks for: **rapid clotting at the wound surface and easy, non-traumatic detachment of the base layer after clot formation**. ⁷⁶

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